

Direction-Preserving Number Representations

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Abstract—Low-precision number formats are widely used in modern machine learning systems due to their efficiency. Accurate direction representation is key to the accuracy of vector representations. This work explores the extent to which the direction of a vector can be represented by selecting its scalar elements from a common finite alphabet of a given size. A geometric framework is introduced for analyzing the directional coverage of such product-structured codes.

This work quantifies the suboptimality gap of such product-structured codes to spherical codes, both in low and in asymptotically high dimensions. Furthermore, within the product code class, standard practice of two’s complement, fixed-point and floating-point are shown to be suboptimal, also with quantified gap, pointing to the potential for new scalar number formats.

Experimentally, such scalar alphabets are numerically optimized across multiple block dimensions for directional coverage, including for the dimension used in NVIDIA’s NVFP4 format. We find that for four bits, their choice of E2M1 closely approximates the optimized alphabet, providing a geometric explanation for its strong empirical performance in low-precision machine learning workloads and an analytical understanding of the link between that superiority and block dimension.

EXTENDED SUMMARY

The increasing computational demands of modern machine learning systems have driven widespread adoption of low-precision numerical representations during both inference and training [1]. To enable low-precision representation, per-element scalar quantization is commonly used in which real-valued vectors are mapped elementwise to a scalar alphabet, resulting in a product code of quantized elements. When evaluating these quantized vectors, most work focuses on the mean-squared error or other norm-based metrics [2]. However, the new industrial dominance of block-based number representations such as block minifloat [3], MX [4], and IEEE3019 [5], where scaling of a (sub)vector is

recorded separately from the vector itself, suggests an alternative metric: directional fidelity.

This observation motivates a shift in perspective for the design of number systems themselves: we ask whether *scalar representations can be designed to preserve vector directions when those scalars populate the elements of a vector*.

We offer the following contributions in this work:

- 1) An analytical proof of the angular error gap between such product-structured codes and optimal spherical codes.
- 2) An analytical proof that among all product codes for a given bitwidth, two’s complement, fixed-point, and floating-point are suboptimal, with quantified suboptimality gap.
- 3) A numerical optimization approach to obtain a direction-preserving alphabet, and an empirical evaluation of contemporary 4-bit formats against this benchmark.
- 4) Open-sourced proofs in Lean, optimization software, and the optimized scalar number formats we find.

A. Direction Preservation in Low Precision

Recent work has increasingly highlighted the importance of preserving directional information in low-precision machine learning workloads. Several works explicitly incorporate direction or inner-product preservation into the quantization objective. TurboQuant [6] includes a secondary optimization stage that directly minimizes inner-product error between original and quantized vectors. PolarQuant [7] parametrizes vectors in polar coordinates and quantizes magnitude and direction separately to better preserve angular structure, similarly to traditional shape-gain quantizers [8], [9].

These approaches focus on designing structured non-product quantization schemes that preserve directional properties. We complement this by analytical characterization and study of the representational limits imposed by a common fixed class of quantizers, namely those derived from a product structure of scalar alphabets. We are able to precisely separate the extent to which directional fidelity (or its loss) is a result of the *particular*

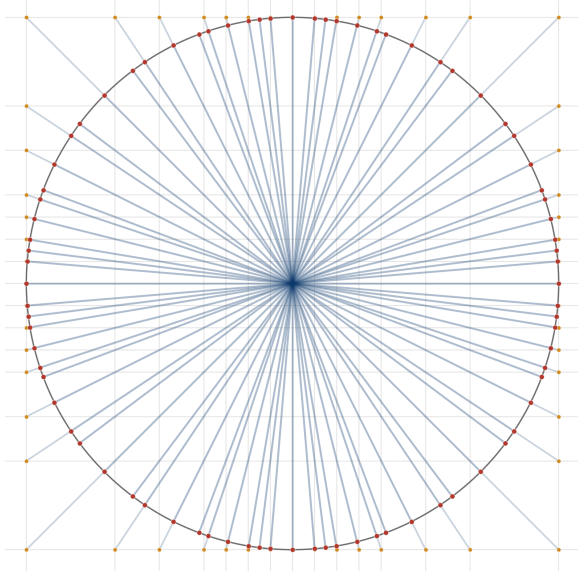


Fig. 1. The directional coverage for a product code of two elements in the 4-bit E2M1 format. Each intersection of the grid lines corresponds to a representable direction through the product structure, projected onto the circle as a red dot. It can be seen that the red dots are not equally spaced in angle, leading to gaps in angle coverage.

number representations (e.g. floating point) versus the product structure itself.

Fundamental Limitations of Product Codes

Product structure imposes intrinsic limitations on directional coverage. Figure 1 shows the coverage of a product code formed by two 4-bit E2M1 elements¹. In the full paper, we will prove a strict quantified separation in directional coverage between product codes and spherical codes.

In two dimensions, a complete classification is provided: aside from the antipodal binary alphabet $\{-a, a\}$, all product codes are strictly worse than spherical codes with the same number of directions.

Extending this to higher dimensions, we can derive lower bounds on the worst-case angular error for any product code through explicit construction of a ‘hard to cover’ witness direction. Combined with classical spherical covering results [11], an asymptotic separation is proven: for any fixed alphabet size, spherical codes strictly outperform product codes in sufficiently high dimensions.

Separation Within Product Codes

Within the product code class, choice of scalar alphabet directly impacts angular coverage.

Formally, for a finite scalar alphabet $A \subset \mathbb{R}$, let

$$\mathcal{P}_n(A) := \{x/\|x\|_2 : x \in A^n \setminus \{0\}\} \subset \mathbb{S}^{n-1}$$

¹E2M1 refers to a floating-point format with exponent width 2 and mantissa width 1. This format is used in NVFP4 [10]. E2M1 does not support infinities or NaN values, and is part of the OCP specification [4].

be the set of representable directions, where $\mathbb{S}^{n-1}$ denotes the $(n-1)$ -dimensional sphere embedded in ambient n -dimensional space. Let $F_n(A)$ be its spherical covering radius:

$$F_n(A) := \sup_{u \in \mathbb{S}^{n-1}} \min_{c \in \mathcal{P}_n(A)} \angle(u, c),$$

where $\angle(\cdot, \cdot)$ denotes angular distance. In the full paper, it will be shown that for fixed alphabet size, $F_n(A) \rightarrow \pi/2$ uniformly as $n \rightarrow \infty$, so raw angular separation is not the most informative asymptotic scale. Thus we study a normalized orthogonality deficit

$$\sqrt{H_n} \cos F_n(A), \text{ where } H_n = \sum_{i=1}^n \frac{1}{i}.$$

We will prove that, for every bit-width $b \geq 3$, optimal arbitrary scalar alphabets asymptotically outperform the best b -bit standard alphabets on this normalized scale.

More precisely, if

$$w_{n,2^b} := \inf\{F_n(A) : |A| = 2^b\}$$

and $F_{n,b}$ denotes the best covering radius among the b -bit two’s complement, fixed-point, and floating-point alphabets with subnormals and no infinities, then the paper will show

$$\liminf_{n \rightarrow \infty} \sqrt{H_n} \cos w_{n,2^b} \geq 2\sqrt{2^{b-1} - 1},$$

whereas

$$\limsup_{n \rightarrow \infty} \sqrt{H_n} \cos F_{n,b} \leq 2\sqrt{\frac{2^{b-1} + 1}{3}}.$$

Since the first constant is strictly larger than the second for $b \geq 3$, the best arbitrary scalar alphabets converge to $\pi/2$ strictly more slowly than the standard alphabets, by at least an explicit constant factor on the $1/\sqrt{\log n} \sim 1/\sqrt{H_n}$ scale.

Therefore, standard formats are suboptimal in direction preservation even within the product code class and when constrained to the same bit budget, thus motivating exploration into new scalar formats.

Empirical Evaluation and Optimized Alphabets

Empirical evaluations are conducted to compare product codes. Directional coverage is evaluated in dimensions up to $d = 64$ using worst-case angular error estimation via sampling on the sphere and optimized nearest-codeword search.

For each dimension, a custom scalar alphabet is optimized using a hybrid global-local strategy of Differential Evolution [12] for global search followed by Powell’s Method [13] for local refinement.

The optimized alphabet is compared against several standard 4-bit formats, including E2M1 as used in NVFP4 [10], E1M2, INT4, and E3M0 (powers-of-two) representations. These results are

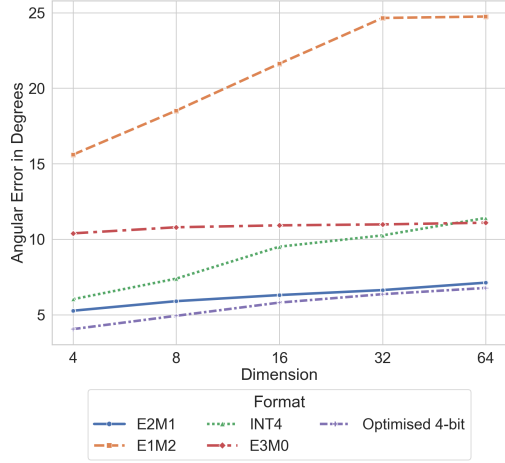


Fig. 2. Sampled worst-case angular error with increasing dimension. The proposed optimized 4-bit alphabet improves on the other number representations, but comes close to E2M1, the format used in NVFP4.

TABLE I
SAMPLED WORST-CASE ANGULAR ERROR FOR 4-BIT ALPHABETS ACROSS VARIOUS DIMENSIONS (DEGREES)

Datatype	$d = 4$	$d = 8$	$d = 16$	$d = 32$	$d = 64$
Optimized Alphabet	4.05	4.93	5.80	6.36	6.77
E2M1	5.26	5.90	6.30	6.63	7.12
INT4	6.03	7.39	9.51	10.3	11.4
E3M0	10.4	10.8	11.0	11.1	11.1
E1M2	15.6	18.5	21.6	24.6	24.7

depicted in Figure 2, and recorded in Table I. The optimized alphabet consistently achieves the lowest angular error. However, the E2M1 format has very similar coverage capabilities, significantly outperforming alternative common formats.

Figure 3 shows 16D log-log regressions between the optimized alphabet and the various formats when the slope is fixed to one (corresponding to directional isometry, as we will prove in the full paper). E2M1 aligns very closely with the optimized alphabet in log-space. E2M1 can thus be interpreted as a specific rounding of the optimized alphabet we propose to the best-fitting 4-bit floating-point family, trading a small reduction in directional fidelity for an increased efficiency in computation.

Conclusion

New theoretical and practical results are established on the limitations of product-structured representations both against spherical codes and within the product code class and a new scalar code is designed.

This work establishes directional preservation as a key principle for future number-system design in low-precision machine learning and computer arithmetic, laying down a theoretical and empirical basis for this study.

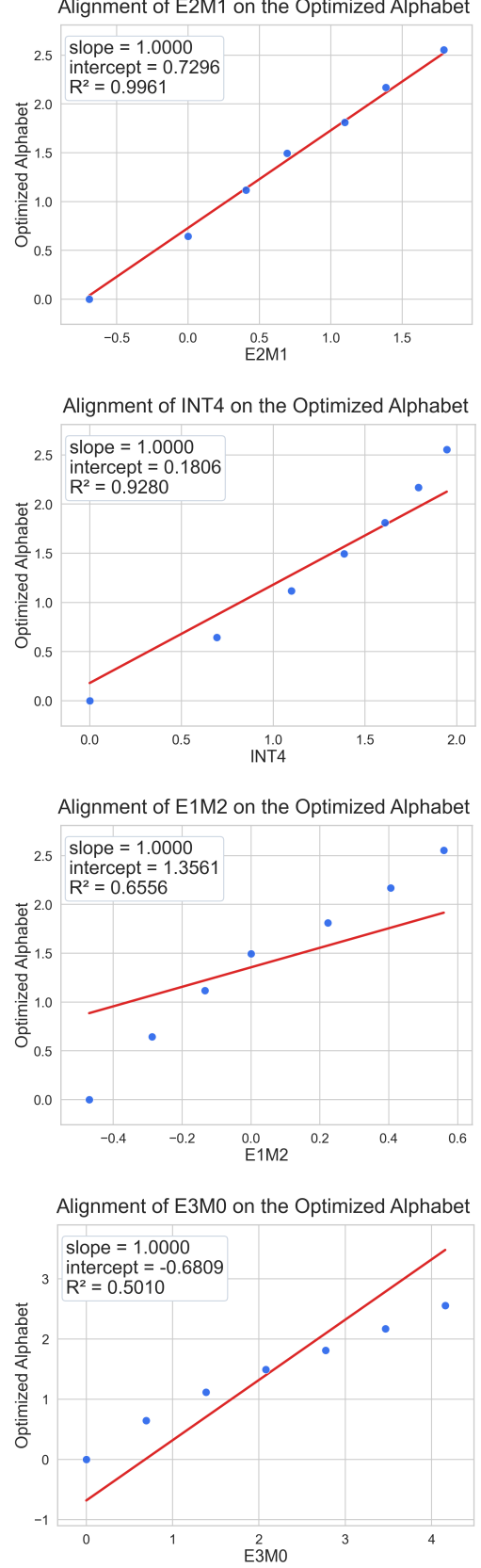


Fig. 3. Log-log regressions of various alphabets versus our optimized alphabet for 16D. The plots show that E2M1 has very strong alignment corresponding to a near isometry with our optimized code. The data point labels show the standard code's representable values.

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